

Lecture-6
Mathematics 1 (15B11MA111)

CO [C105.1]

Module: Partial Differentiation

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics”
fifth edition, Narosa publishing house, 2016.

Topics to be covered

- Homogeneous Function of Degree n
- Euler's Theorem for Homogeneous Functions
- Verification of Euler's Theorem
- Proof of Euler's Theorem
- Some Examples
- Euler's Theorem of second order
- Exercise

Homogeneous Functions

Definition: A function, in which every term is of the n^{th} degree, is called a homogeneous function of degree n .

A homogeneous function can be written as

$$f(x, y) = a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \dots + a_n y^n \quad (1)$$

where $a_0, a_1, a_2, \dots, a_n$ are constants.

If λ is any positive number, then replacing x by λx and y by λy in (1), we observe that

$$\begin{aligned} f(\lambda x, \lambda y) &= a_0 (\lambda x)^n + a_1 (\lambda x)^{n-1} \lambda y + a_2 (\lambda x)^{n-2} (\lambda y)^2 + \dots + a_n (\lambda y)^n \\ &= \lambda^n (a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \dots + a_n y^n) \\ &= \lambda^n f(x, y) \end{aligned}$$

The expression (1) may also be written as

$$x^n \left[a_0 + a_1 \frac{y}{x} + a_2 \left(\frac{y}{x} \right)^2 + \dots + a_n \left(\frac{y}{x} \right)^n \right] = x^n f\left(\frac{y}{x} \right)$$

Thus a function which satisfies $f(\lambda x, \lambda y) = \lambda^n f(x, y)$ for any positive number λ , or can be expressed as $x^n \phi(y/x)$ is a **homogeneous function of degree n** .

Example: Check whether the following function is homogeneous if yes then find its degree

$$f(x, y) = x^3 + 7xy^2$$

Solution:

$$\begin{aligned} f(\lambda x, \lambda y) &= (\lambda x)^3 + 7(\lambda x)(\lambda y)^2 \\ &= \lambda^3 (x^3 + 7xy^2) \\ &= \lambda^3 f(x, y) \end{aligned}$$

OR

$$f(x, y) = x^3 \left(1 + 7 \left(\frac{y}{x} \right)^2 \right) = x^3 \phi \left(\frac{y}{x} \right)$$

Homogeneous Degree

Yes

3

Examples:

Check whether the following functions are homogeneous if yes then find its degree

	Homogeneous	Degree
$f(x, y) = 3x^2y^2 - 5x^3y$	Yes	4
$f(x, y) = 3x^2y^2 - 5xy$	No	
$f(x, y) = \frac{3x^2y^2 - 5x^3y}{\sqrt{x+8y}}$	Yes	7/2
$f(x, y) = xy \sin^{-1}\left(\frac{y}{x+y}\right)$	Yes	2
$f(x, y) = xy + \sin^{-1}\left(\frac{y}{x+y}\right)$	No	
$f(x, y) = \sin^{-1}\left(\frac{x}{\sqrt{x+y}}\right)$	No	

Function of 3 variables

$f(x, y, z) = \frac{\sqrt{4x^5 + 2y^2z^3}}{x - 5y + 3z}$	Yes	3/2
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Euler's Theorem:

If $f(x, y)$ is a homogeneous function of degree n , then

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f$$

Converse is also true.

Exercise: Verify Euler's theorem for the functions

(i) $f(x, y) = x^3 + 7xy^2$ $f(x, y, z) = 3x^2yz + 5xy^2z - 4z^4$

Solution: (i) The given function is a homogeneous function of degree 3.

$$f_x(x, y) = 3x^2 + 7y^2 \quad f_y(x, y) = 14xy$$

$$\begin{aligned} xf_x + yf_y &= x(3x^2 + 7y^2) + y(14xy) \\ &= 3x^3 + 7xy^2 + 14xy^2 \\ &= 3(x^3 + 7xy^2) \end{aligned}$$

$$\therefore xf_x + yf_y = 3f(x, y)$$

Exercise Cont...: Verify Euler's theorem for the functions

(i) $f(x, y) = x^3 + 7xy^2$ $f(x, y, z) = 3x^2yz + 5xy^2z - 4z^4$

Solution (ii): The given function is a homogeneous function of degree 4.

$$f_x(x, y, z) = 6xyz + 5y^2z$$

$$f_y(x, y, z) = 3x^2z + 10xyz$$

$$f_z(x, y, z) = 3x^2y + 5xy^2 - 16z^3$$

$$\begin{aligned} xf_x + yf_y + zf_z &= x[6xyz + 5y^2z] + y[3x^2z + 10xyz] + z[3x^2y + 5xy^2 - 16z^3] \\ &= 12x^2yz + 20xy^2z - 16z^4 \\ &= 4[3x^2yz + 5xy^2z - 4z^4] \end{aligned}$$

$$\therefore xf_x + yf_y + zf_z = 4f(x, y, z)$$

Proof of Euler's Theorem: If $f(x, y)$ is a homogeneous function of degree n , then

$$f(x, y) = x^n \phi\left(\frac{y}{x}\right)$$

$$f_x = nx^{n-1} \phi\left(\frac{y}{x}\right) + x^n \phi'\left(\frac{y}{x}\right) \left(\frac{-y}{x^2}\right)$$

$$f_y = \left[x^n \phi'\left(\frac{y}{x}\right) \right] \left(\frac{1}{x}\right)$$

$$\begin{aligned} xf_x + yf_y &= x \left\{ nx^{n-1} \phi\left(\frac{y}{x}\right) - x^{n-2} y \phi'\left(\frac{y}{x}\right) \right\} + y \left\{ x^{n-1} \phi'\left(\frac{y}{x}\right) \right\} \\ &= nx^n \phi\left(\frac{y}{x}\right) - x^{n-1} y \phi'\left(\frac{y}{x}\right) + x^{n-1} y \phi'\left(\frac{y}{x}\right) \\ &= nx^n \phi\left(\frac{y}{x}\right) = nf(x, y) \end{aligned}$$

Hence

$$xf_x + yf_y = nf(x, y)$$

Question: If $u = \sin^{-1} \left(\frac{x^2 y^3}{x+y} \right)$, that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3 \tan u.$$

Solution: The function $u = \sin^{-1} \left(\frac{x^2 y^3}{x+y} \right)$ is not homogeneous.

Let $\sin u = \frac{x^2 y^3}{x+y} = f(x, y)$

Since $\sin u = \frac{x^2 y^3}{x+y} = x^4 \frac{\left(\frac{y}{x}\right)^3}{1 + \frac{y}{x}} = x^4 \phi \left(\frac{y}{x} \right)$

Therefore $\sin u$ is a homogeneous function of degree 4.

Using Euler's theorem $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 4f$

$$x \frac{\partial(\sin u)}{\partial x} + y \frac{\partial(\sin u)}{\partial y} = 4 \sin u$$

$$x \cos u \frac{\partial u}{\partial x} + y \cos u \frac{\partial u}{\partial y} = 4 \sin u$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 4 \tan u$$

Exercise:

If $u(x, y)$ is a homogeneous function of degree n then show that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u.$$

Proof: Since u is a homogeneous function of degree n then by Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad (1)$$

Differentiating partially to equ. (1) w. r. to x

$$\frac{\partial u}{\partial x} + x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} = n \frac{\partial u}{\partial x} \quad (2)$$

Differentiating partially to equ. (1) w. r. to y

$$x \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial y} + y \frac{\partial^2 u}{\partial y^2} = n \frac{\partial u}{\partial y} \quad (3)$$

Multiplying (2) by x and (3) by y and adding

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right)$$

$$nu + x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} \quad (\text{Using Euler's Theorem})$$

Hence

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

Exercise:

- ❖ With and without using Euler's theorem find $xf_x + yf_y$ for

$$f(x, y) = \tan^{-1}\left(\frac{xy}{x^2 + y^2}\right) \quad \text{Ans. } 0$$

- ❖ Use Euler's theorem to find $xf_x + yf_y$ for

$$f(x, y) = \frac{x+y}{x^2 + y^2} + \frac{xy}{x^4 + y^4} \sin^{-1}\left(\frac{y}{x}\right) \quad \text{Ans. } -\frac{x+y}{x^2 + y^2} - 2\frac{xy}{x^4 + y^4} \sin^{-1}\left(\frac{y}{x}\right)$$

- ❖ Using Euler's theorem find $xf_x + yf_y$ and $x^2f_{xx} + 2xyf_{xy} + y^2f_{yy}$ for

$$f(x, y) = \sin^{-1}\left(\frac{x+y}{\sqrt{x} + \sqrt{y}}\right) \quad \text{Ans. } \frac{1}{2} \tan f, \quad \frac{1}{2} \tan f \left(\frac{1}{2} \sec^2 f - 1 \right)$$

Thank You